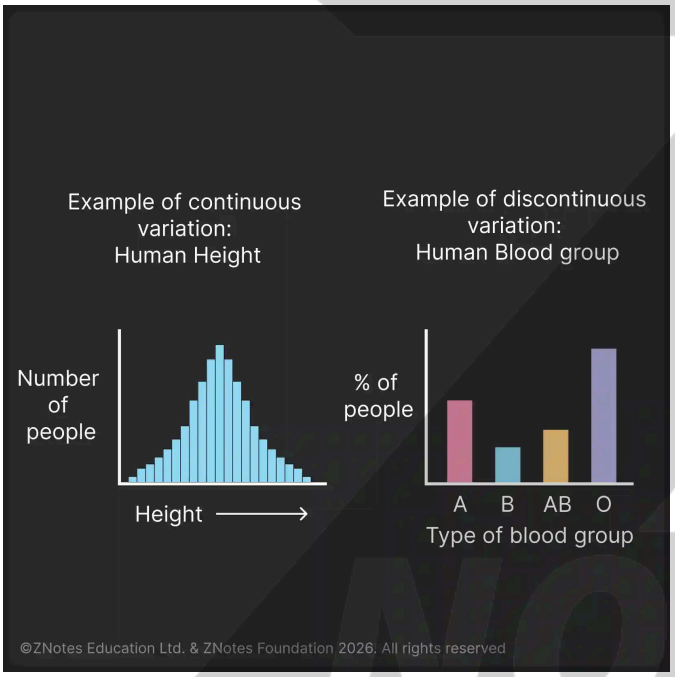
1. Variation and Selection

1.1. Variation

- **Variation:** differences between individuals of the same species
- Both genetic and environmental factors cause phenotypic variation
- **Continuous Variation:** results in a range of phenotypes between two extremes; examples include body length and body mass
- **Discontinuous Variation:** results in a limited number of phenotypes with no intermediates (e.g. ABO blood groups, seed shape in peas and seed colour in peas)
- **Continuous variation** is caused by genes and the environment
- **Discontinuous variation** is usually caused only by genes

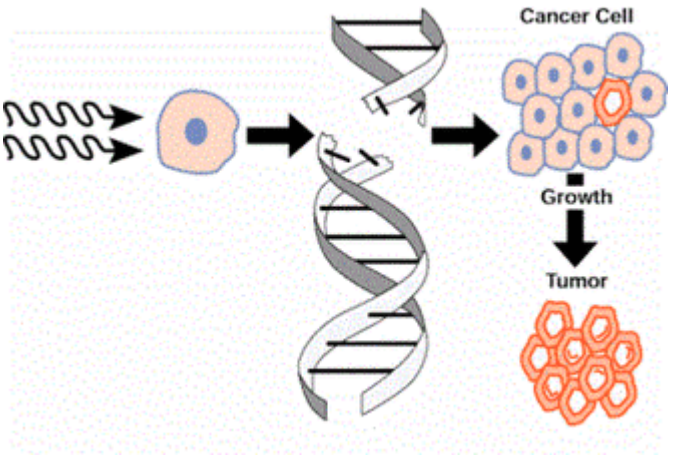


Syllabus 18.1.5: You must be able to investigate and describe examples of continuous and discontinuous variation

Mutation

Mutation: A genetic change.

- **Gene Mutation:** a change in the base sequence of DNA
- Mutation is the way in which new alleles are formed
- **Mutation, meiosis, random mating and random fertilisation** are sources of genetic variation in populations
- Ionising radiation and some chemicals increase the rate of mutation



1.2. Adaptive Features

Adaptive Feature: an inherited (structural) feature that helps an organism to survive and reproduce in its environment

- **Xerophytes** live in deserts where water is scarce, and evaporation is rapid or in windy habitats. Their features are:
  - Deep roots reach the water far underground
  - Leaves have reduced spines with minimum surface area for transpiration
  - Shallow spreading roots to collect occasional rain
  - Rolled leaves, leaf hairs, and stomata sunk in pits to trap moist air
  - Waxy leaf cuticle, impermeable to water
  - Stomata open at night and close at midday when evaporation is highest
  - E.g. cactus and marram grass
- **Hydrophytes:** live wholly or partly submerged in water. Their features are:
  - Leaves are highly divided to create a large surface area for absorption and photosynthesis
  - Minimal cuticle formation
  - Lack of xylem tubes, no stomata underside of leaves
  - Stomata are on the upper surface and have a thick waxy layer to repel water and to keep the stomata open and clear
  - Roots are often reduced, and root hairs are often absent
  - Aerenchyma tissues (air filled tissues) are present to help the plant stay afloat.

1.3. Selection

Natural Selection

In any environment, the individuals that have the best adaptive features are the ones most likely to survive and reproduce.

- The greater chance of the best-adapted organisms passing on genes.

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- The development of strains of antibiotic-resistant bacteria is an example of natural selection.
- The surviving organisms reproduce since they don't get eaten up, so variation has caused the species to evolve.

Process of Natural Selection:

1. genetic variation within populations
2. production of many offspring
3. struggle for survival, including competition for resources
4. Individuals better adapted to the environment have a greater chance of reproduction than others; these individuals pass on their alleles to the next generation.

Adaptation: the process of natural selection by which populations become more suited to their environment over many generations.

Artificial Selection

Artificial Selection: breeds organisms with valued characteristics together to produce offspring that share those valuable characteristics.

- It can be used to produce organisms that are more economically valued
- For example, cows that produce more milk, wheat that is easier to separate from grain, dogs that have a better appearance

Process of Selective Breeding:

- Selecting by individuals with desirable features
- Crossing these individuals to produce the next generation
- Selection of offspring showing the desirable features

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